

Definition 1. Bellman Equation. Let the state evolve according to $x_{t+1} = f(x_t, a_t)$, and let per-period payoff be $u(x_t, a_t)$. The value function $V(x)$ satisfies the Bellman equation

$$V(x) = \max_{a \in A(x)} \{u(x, a) + \beta V(f(x, a))\}.$$

For example, for a production function $f(k)$, utility $u(x)$, we have

$$V(x) = \max_{k'} \{u(f(k) - k') + \beta V(k')\}.$$

With uncertainty, we have

$$V(k, A) = \max_{k'} \{u(f(k, A) - k') + \beta \mathbb{E}V(k', A')\}$$

FOC for Bellman Equation: The first-order condition for the control a is

$$\frac{\partial u(x, a^*)}{\partial a} + \beta \frac{\partial V(f(x, a^*))}{\partial x} \frac{\partial f(x, a^*)}{\partial a} = 0.$$

Envelope Condition for Bellman Equation: The envelope condition is

$$\frac{\partial V(x)}{\partial x} = \frac{\partial u(x, a^*)}{\partial x} + \beta \frac{\partial V(f(x, a^*))}{\partial x} \frac{\partial f(x, a^*)}{\partial x}.$$

1 Diff Eqs

Integrating Factor: Start with the linear ODE

$$\dot{y}_t + a_t y_t = b_t,$$

and define the integrating factor

$$\mu_t = e^{\int a_t dt}.$$

Multiply the ODE by μ_t :

$$\mu_t \dot{y}_t + a_t \mu_t y_t = \mu_t b_t,$$

so the left-hand side becomes

$$\frac{d}{dt}[\mu_t y_t] = \mu_t b_t.$$

Integrate in time and solve for y_t :

$$\mu_t y_t = \int \mu_t b_t dt + C.$$

2 Hamiltonian

Hamiltonian: sum instantaneous utility + shadow value times the law of motion, and call it a day.

$$H(c_t, k_t, \mu_t) = \underbrace{u(c_t)}_{\text{instantaneous utility}} + \overset{\text{shadow value}}{\tilde{\mu}_t} \underbrace{f(k_t, c_t)}_{\text{law of motion}}$$

- Integrating factor for ODEs
- ODEs: log trick

Optimal Control Problem: Three conditions:

$$\text{Optimality: } \frac{\partial H}{\partial c} = 0, \quad \text{Multiplier Condition: } \frac{\partial H}{\partial k} = \rho \mu_t - \dot{\mu}_t, \quad \text{Hamiltonian: } \dot{k}_t = \frac{\partial H}{\partial \mu}$$

3 HJB

HJB:

$$\rho v(k) = \max_c \{u(c) + [F(k) - \delta k - c]v'(k)\} \quad (1)$$

$$\frac{\partial}{\partial c} [u(c) + (F(k) - \delta k - c)v'(k)] = 0 \implies u'(c) = v'(k). \quad (2)$$

Ito's Lemma:

Let $(X_t)_{t \geq 0}$ satisfy

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t,$$

and let $f : [0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$ be $C^{1,2}$. Then Itô's lemma gives

$$df(t, X_t) = \left(\frac{\partial f}{\partial t}(t, X_t) + \mu(t, X_t) \frac{\partial f}{\partial x}(t, X_t) + \frac{1}{2} \sigma^2(t, X_t) \frac{\partial^2 f}{\partial x^2}(t, X_t) \right) dt + \sigma(t, X_t) \frac{\partial f}{\partial x}(t, X_t) dW_t.$$

HJB with stochastic processes: